VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-IITI-HT-0050

Dermasense: a laser-written strip that reads a dressing's own 'change line'

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Health Tech: Emergency Response & Remote Monitoring

Evidence key: • verified • designed, lab test pending • future scope

Problem Context

PUBLISHED EVIDENCE · CITED

44.9 %	≈ 2 cm	1 point	Uneven
OF 588 CHANGES MADE AT OPTIMAL MOISTURE[1]	MAKER'S 'CHANGE NOW' LINE FROM PAD EDGE[2]	WHERE TODAY'S MOISTURE SENSOR READS[1]	IN-VITRO MOISTURE MAP, 8 POINTS[3]

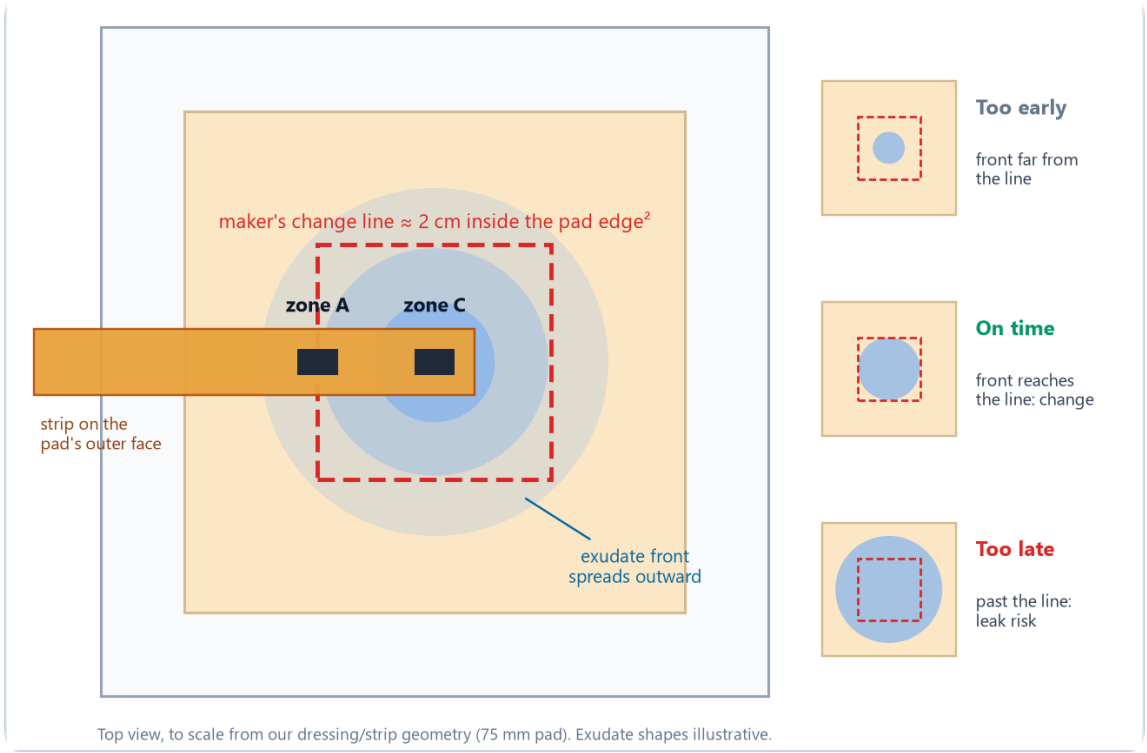


Figure: the change line is invisible under a cover (to scale, our design).

Target Beneficiary & Problem Statement

- **Who:** nurses and home carers of chronic wounds under foam dressings.
- **Pain:** the maker's 'change now' sign is visual[14], and the dressing is covered.
- **Today:** change by the clock, by lifting the cover, or after a leak.
- **Gap:** in our search, no low-cost sensor reads that change line[45].

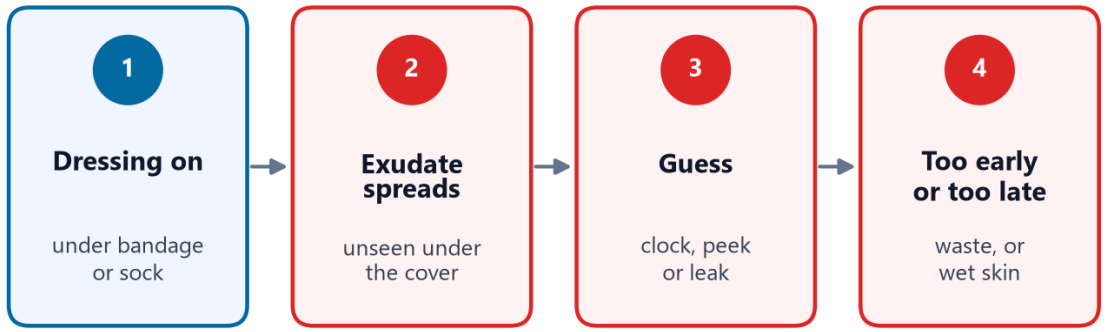
Problem: carers cannot see when a covered dressing is really full.

User Journey: Before State

CURRENT STATE · CITED

44.9 %	≈ 18 min	₹240–510	0.1–1.3 g/cm²·day
CHANGES WHILE MOISTURE WAS STILL OPTIMAL[1]	A DRESSING VISIT, PATIENT-REPORTED[13]	ONE 10 × 10 CM FOAM DRESSING, RETAIL[15]	REPORTED EXUDATE RATES, 2 METHODS[6]

Today: checked by schedule or by looking



44.9 % of changes came while moisture was still optimal[1]

Cited numbers only (References slide)

Figure: today's pain-point map (cited numbers only).

Current Pain Point Mapping

- **Time:** every check means unwrap, look, re-secure: about 18 min a visit[13].
- **Waste:** 44.9 % of changes came while moisture was still optimal[1]; each discards a dressing[15].
- **Risk:** a late change can leak: wet skin, soiled clothing.
- **Cause:** reported exudate rates span over 10×[6]; a clock cannot follow.

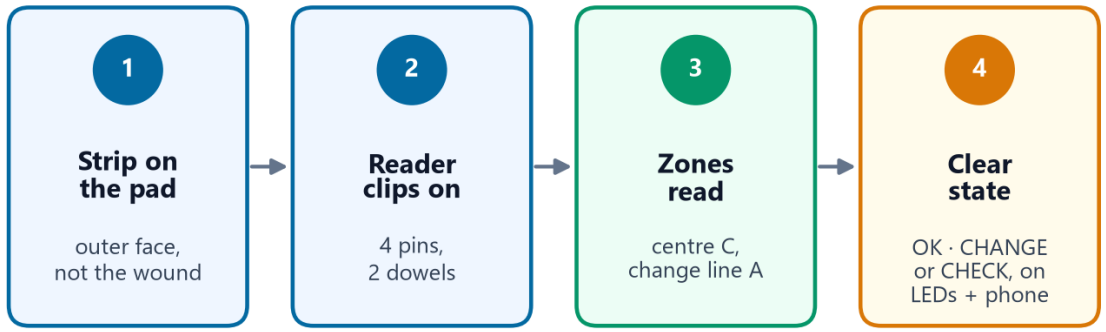
Quantified Inefficiency: published figures only (a UK patient survey, Indian retail prices); we will time real checks with nurses.

User Journey: After State (Intended)

INTENDED · NOT CLINICALLY VALIDATED

up to 44.9 %	≤ ₹100	±10 %	0
EARLY CHANGES A RELIABLE ALERT COULD AVOID[1]	TARGET STRIP COST (ESTIMATE)	TARGET ALERT ACCURACY (BENCH, E4)	CONTACT WITH THE WOUND BED (BY DESIGN)

Intended: the strip reads the change line



Target: alert within ±10 % of the visible front (bench test E4)

INTENDED WORKFLOW · NOT CLINICALLY VALIDATED

Figure: the intended workflow; bench-testable, not clinically validated.

Future Solution Mapping & Quantified Efficiency

- **Same routine:** strip on the pad, reader clips on.
- **Clear signal:** CHANGE when the line is wet; CHECK if it wets first.
- **Remote:** the state is broadcast over Bluetooth to a phone.
- **Payback:** a ≤ ₹100 strip pays for itself if 1 in 2–5 changes is avoided[15].

Design Principle: only film and carbon are disposable; the strip never touches the wound.

System Architecture & Specs

DESIGNED · LAB TEST PENDING

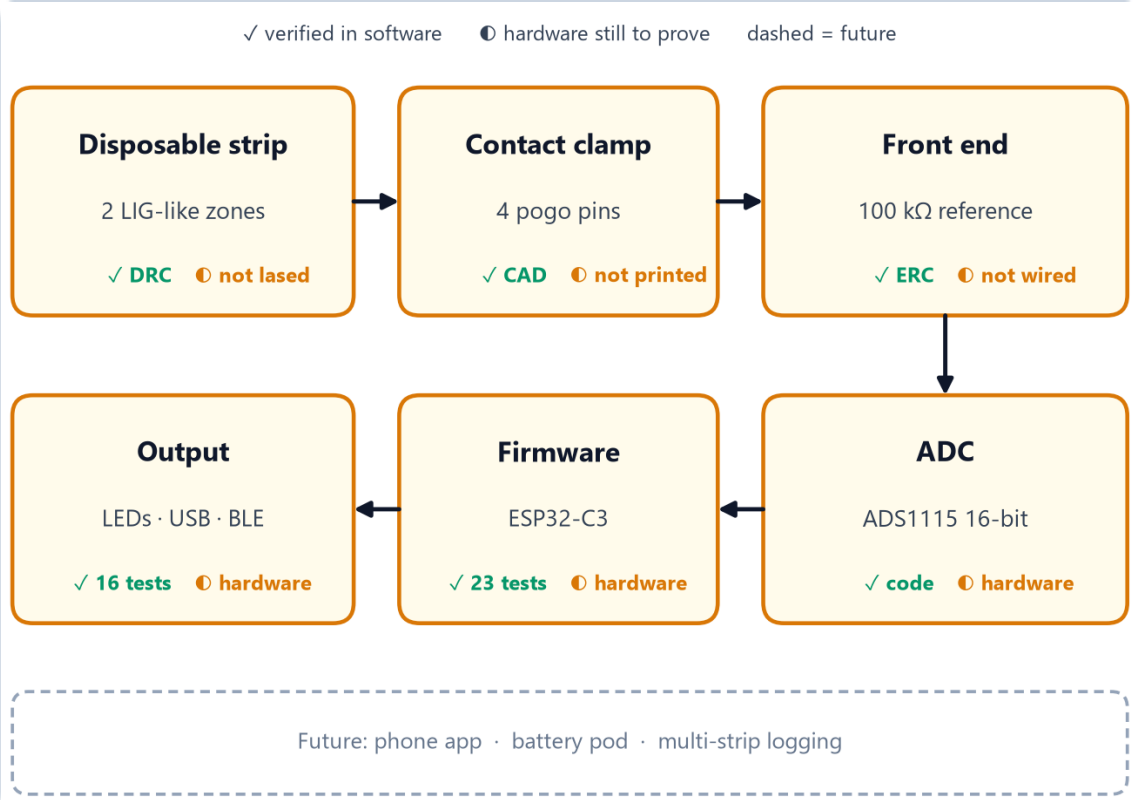


Figure: subsystems; ✓ = verified in software, ◐ = hardware still to prove.

Target Specifications & Hardware

Parameter	Design value
Sensing zones	2: pad centre C, change line A
Wet / dry	< 100 kΩ / > 300 kΩ, 3 scans
Drive	bipolar, ≤ 33 µA, no net DC
ADC / MCU	ADS1115 16-bit (I²C) · ESP32-C3
Contacts	4 pogo pins + 2 dowels
Output	LEDs · USB log · BLE broadcast

Architecture Rationale: only film and carbon are disposable; every electronic part is reused.

System Architecture & Specs: Firmware & Data Flow

VERIFIED IN SOFTWARE · HARDWARE PENDING

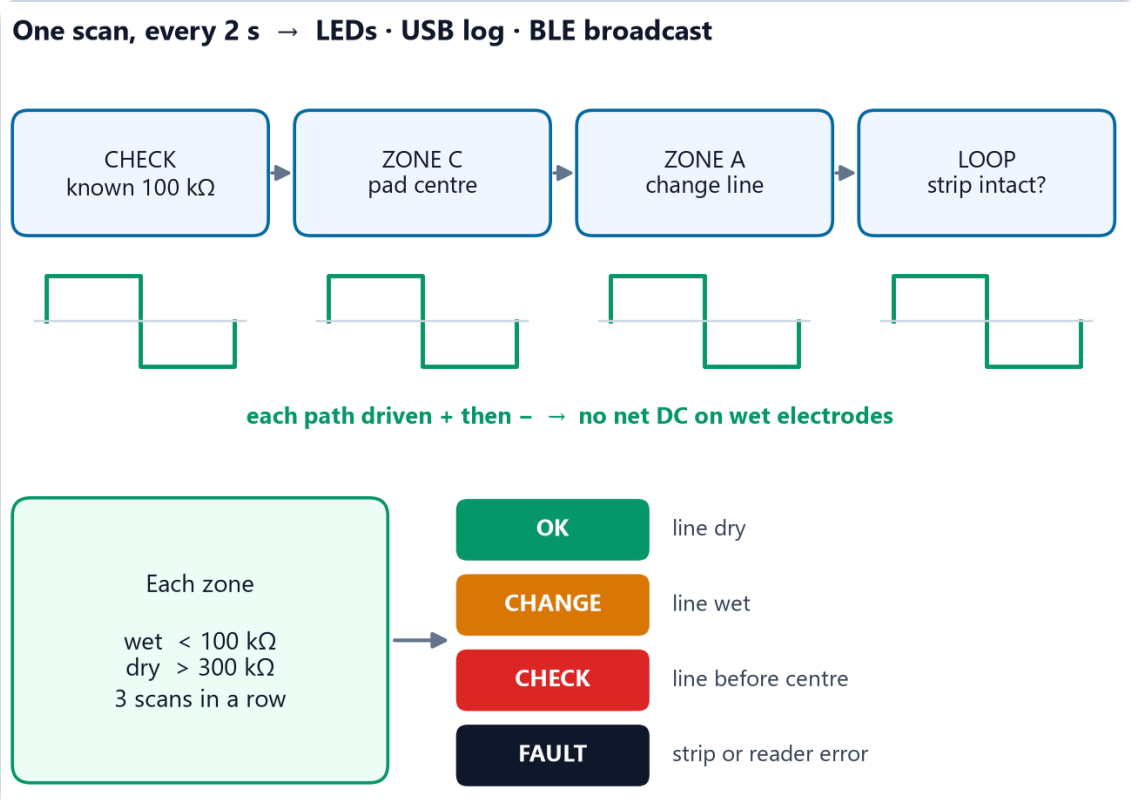


Figure: one reader scan, from measurement to state.

Firmware Logic & Data Flow

- **Measure:** each zone driven both ways, so no net DC.
- **Read:** 16-bit ADC, scaled to the 3V3 rail.
- **Decide:** wet < 100 kΩ, dry > 300 kΩ, 3 scans in a row.
- **Self-check:** a known 100 kΩ every scan, else FAULT.
- **Share:** CSV over USB, and a Bluetooth broadcast to a phone.

Software vs Hardware: 23 compile-time and 16 PC tests pass; not yet run on a wired board.

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT

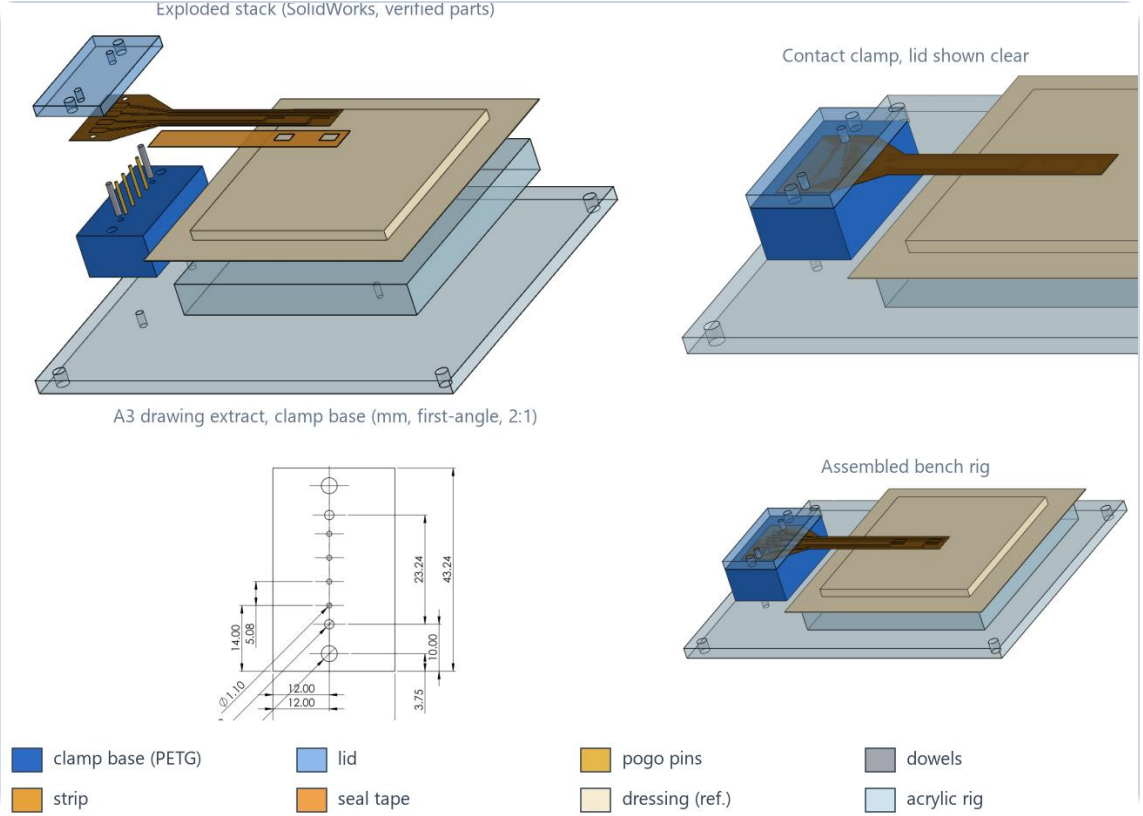


Figure: our SolidWorks 2019 models: exploded stack, clamp, drawing, assembly.

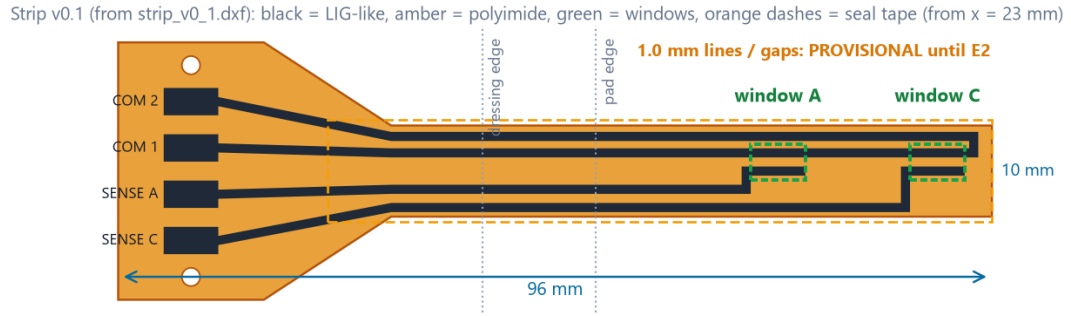
Materials & Mechanical Checks

- **Materials:** polyimide strip · PETG clamp · acrylic rig.
- **Contact:** 4 pogo pins, 0.72 N each (calculated).
- **Alignment:** 2.8× tolerance margin on 3 mm pads.
- **Verified:** 99 automated CAD checks pass, 0 fail.
- **Provisional:** strip sizes wait for E2 laser data.

Design Iteration: a clash check moved the seal tape off the clamp (decision D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · LAB TEST PENDING



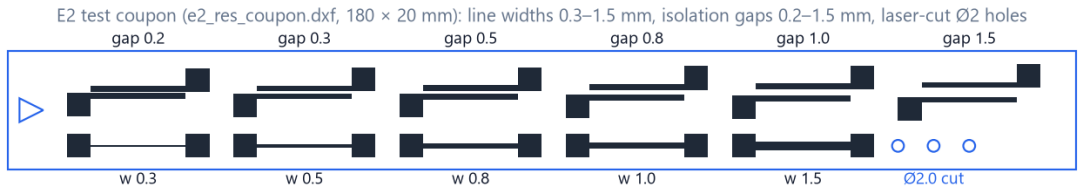


Figure: our laser files, drawn from the DXFs. Not yet lased.

Fabrication & Process Validation (E2)

- **Material:** laser-carbonised, LIG-like layer on polyimide[7].
- **Reference:** setting '12', most likely % PWM, not watts.
- **E2 plan:** 7 coupons, 2 days, 72 pre-registered samples.
- **Pass rules:** CV ≤ 20 %, gaps > 3 MΩ, fixed beforehand.
- **Sealed:** leads under tape; LIG made in air wets easily[9].

Laser Files Verified: 35/35 file checks pass; LaserCAD import is checked at the machine.

Electronics Schematics & BOM

ERC VERIFIED · NOT YET WIRED

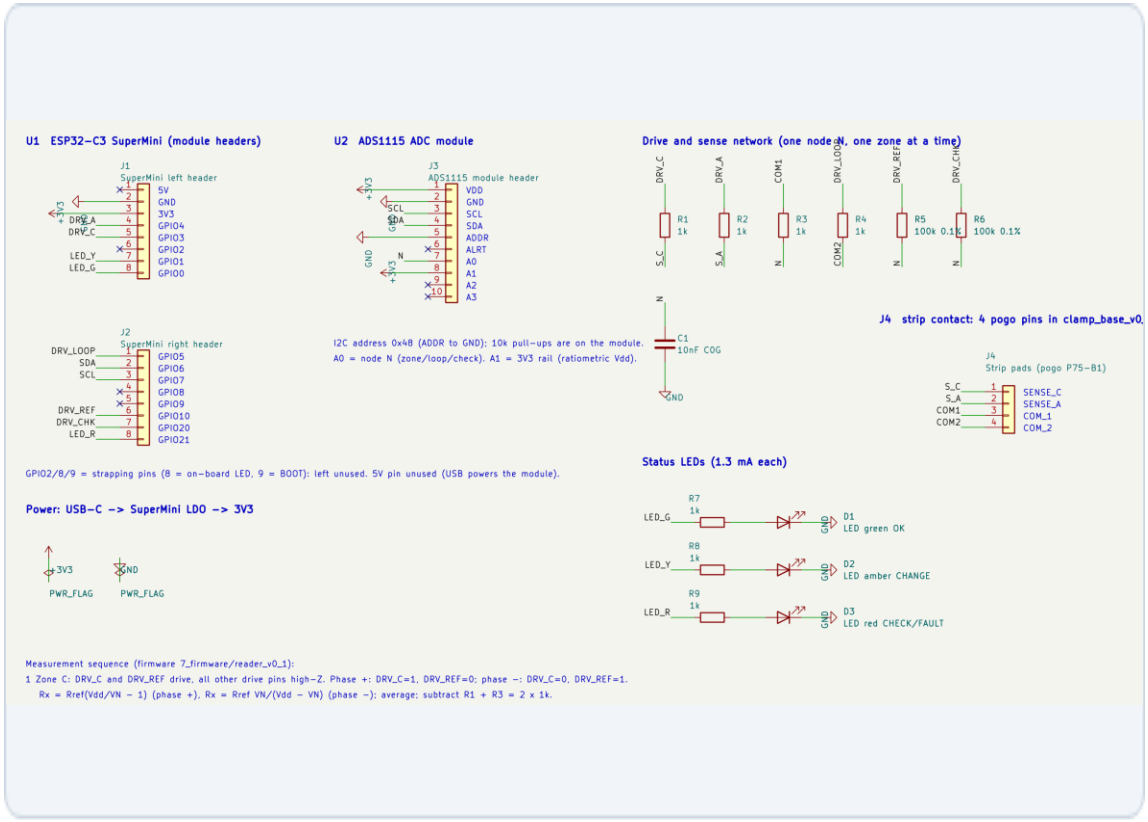


Figure: our KiCad 10 schematic of the reader (exported by kicad-cli).

Key Connections

- **Drive:** GPIO3 / GPIO4 → 1 kΩ → zones C, A.
- **Sense:** node N → 100 kΩ reference → ADC A0.
- **Check:** 100 kΩ on GPIO20; strip loop on GPIO5.
- **Strip:** 4 pogo pins on 3 × 6 mm pads.
- **Power:** USB-C → 3V3 regulator on the module.

Verification: ERC 0 violations; netlist matches firmware and strip pads (0 mismatches).

Electronics Schematics & BOM: Bill of Materials

3 PRICES QUOTED · NOTHING ORDERED

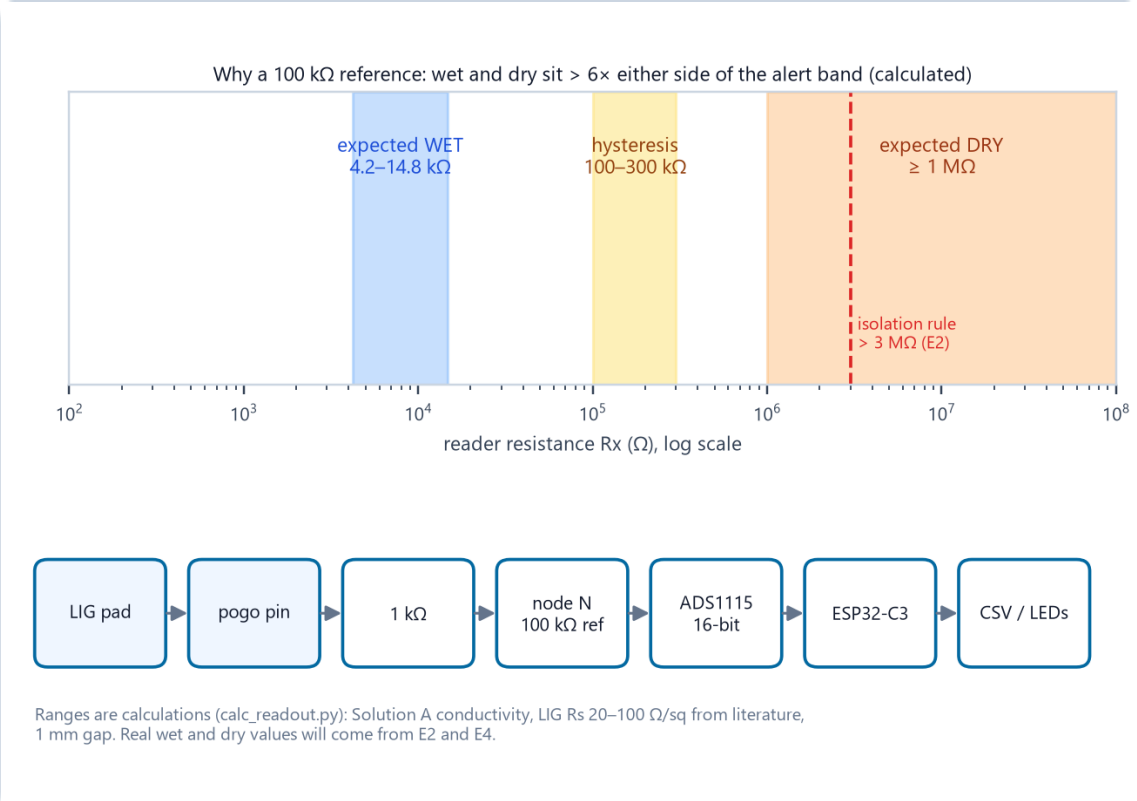


Figure: why a 100 kΩ reference (calculated ranges).

Prototype Bill of Materials

Item	Qty	Source	₹
ESP32-C3 SuperMini	1	Probots · quoted	242
ADS1115 16-bit ADC	1	Probots · quoted	125
P75-B1 pogo pins	100	Hubtronics · quoted	354
Resistors, COG cap, LEDs, board	set	local / online	158–421
Clamp + bench rig	set	lab / local	100–1,240
Strip materials	set	lab / online	800–3,500
Test consumables	set	local	480–1,060
Total			**≈ 2,259–6,942**

Procurement Status: reader parts and pins quoted 19 Oct, in stock in India; the rest are estimates.

Maker Journey

GENUINE ARTIFACTS · NO STAGED EVIDENCE

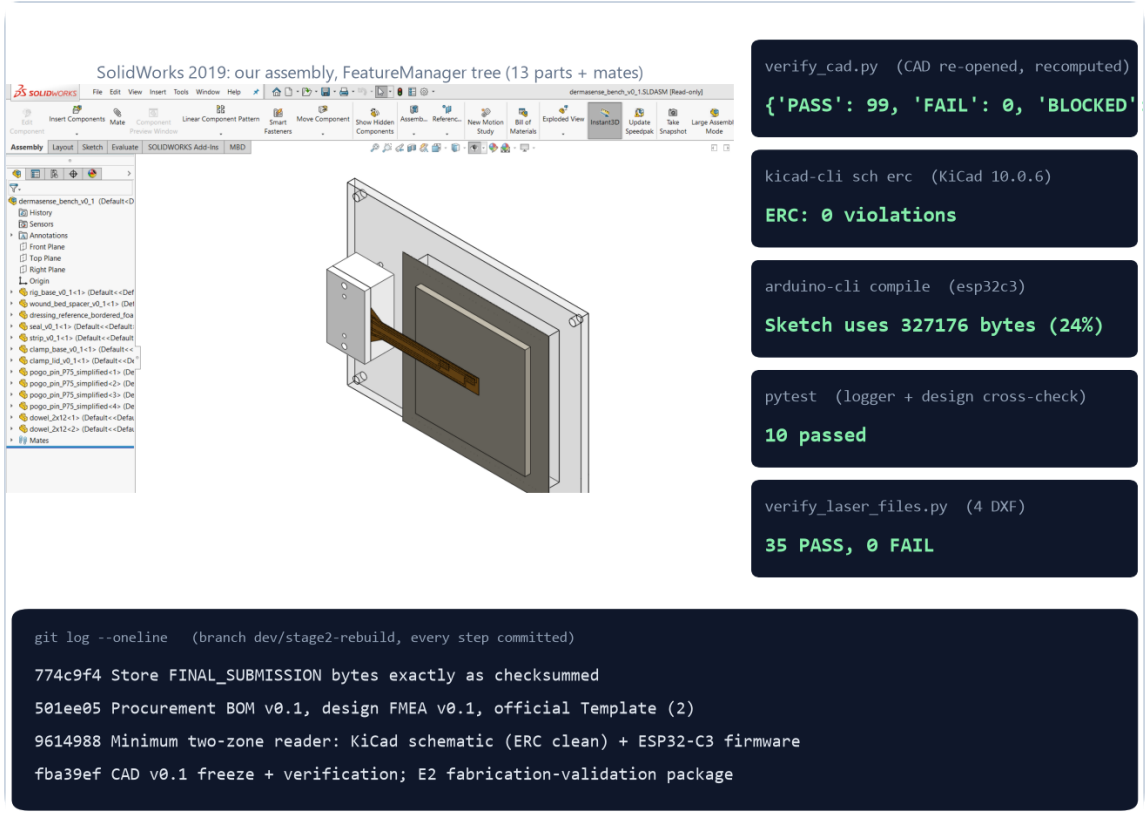


Figure: our SolidWorks window and verbatim tool output (14–15 Oct 2026).

Authenticity: What Each Artifact Proves

- **CAD:** our assembly open in SolidWorks, 13 parts.
- **Checks:** CAD, ERC, compile and tests all re-runnable.
- **History:** every step a git commit since 14 Oct.
- **Photos:** none yet; the first laser session is next.

Raw Repository Linkage: [team to insert the GitHub / Drive link before upload]

Maker Journey: Engineering Iterations

DOCUMENTED DECISIONS

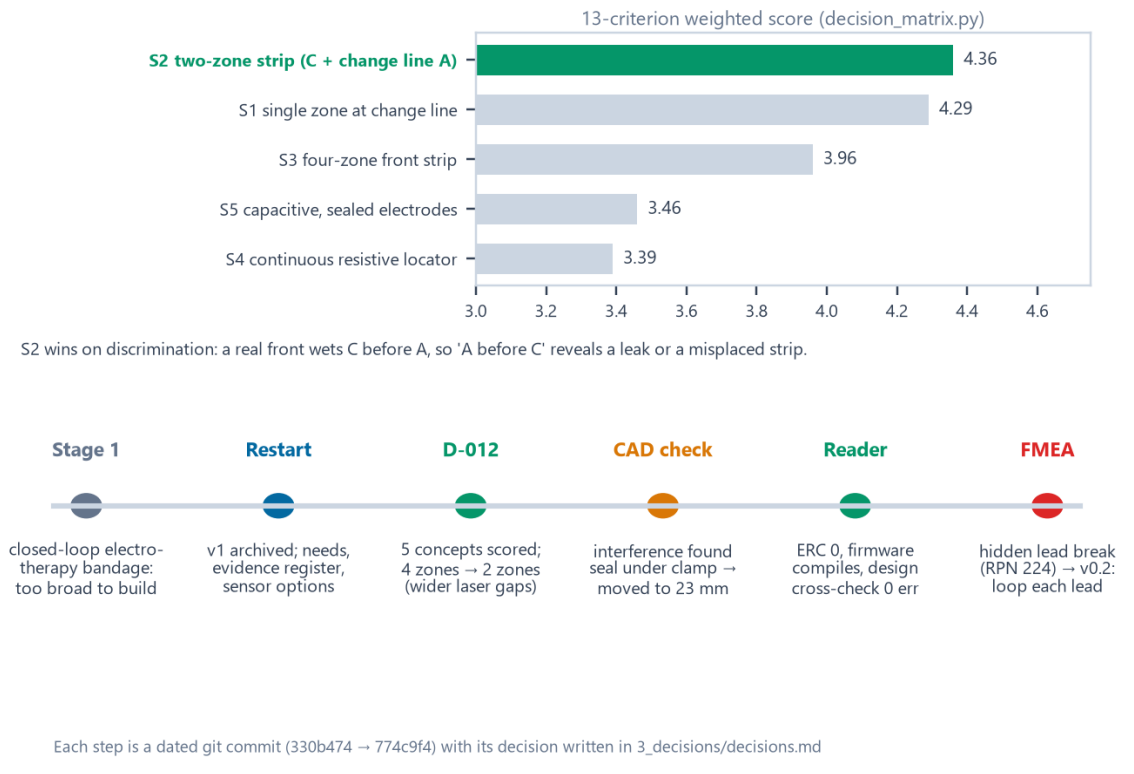


Figure: concept scores and the iteration path (each a dated commit).

Stage 1 to Stage 2 Iterations

- **Pivot:** therapy bandage (already shown in labs[5]) → one job: is it full?
- **Choice:** two-zone strip (4.36) beat one zone (4.29): it spots leaks.
- **Kept:** Stage 1's Bluetooth telemetry, now a state broadcast.
- **Corrected:** Stage 1 called Kapton 'medical-grade'; DuPont cautions otherwise[11].
- **Redesign:** FMEA risk → v0.2 loops each sense lead.

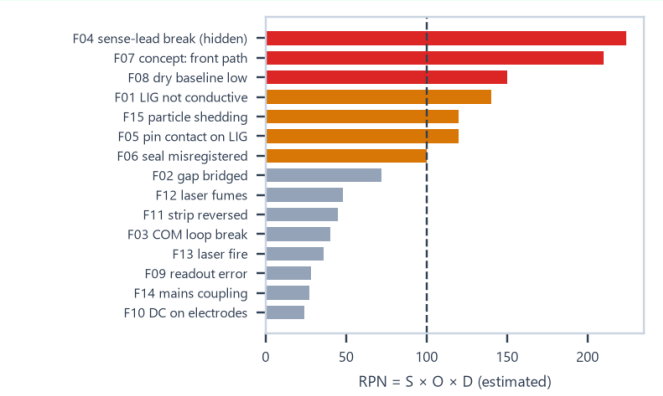
Prior Art: diaper strips[4] and visual change indicators[14] exist; our novelty is reading the change line electrically, with an order check.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

Design FMEA: top risks of 15 (ratings estimated before tests)

Failure mode	RPN	Fix
Cracked sense lead reads 'dry'	224	v0.2: loop each lead to a 2nd pad
Front path differs from model	210	test on real foam dressings (E4)
Humid dressing reads 'wet'	150	measure dry baselines (E4)



Fail-safes

- **Bench only:** USB reader, never on a person.
- **No net DC:** bipolar drive, idle pins off.
- **Self-check:** every scan, else FAULT.
- **No wound contact:** strip on the pad's outer face.

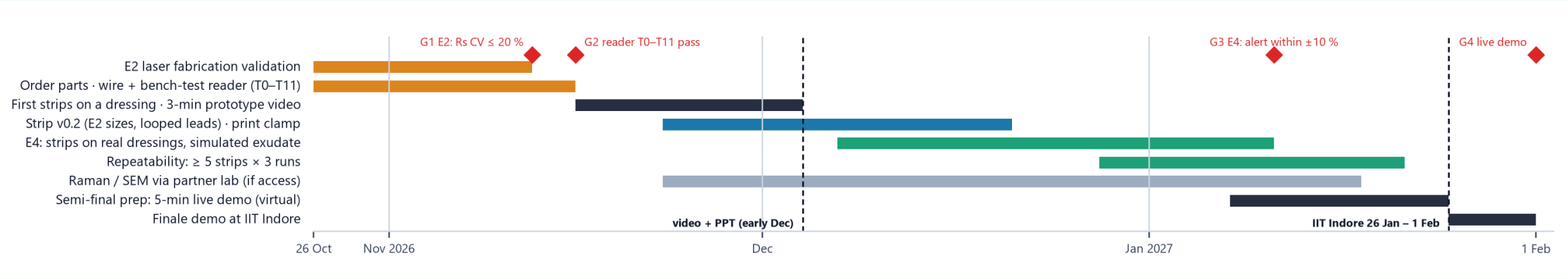
Standards (considered, not claimed)

- **ISO 14971** risk management
- **IEC 60601-1** future on-body reader
- **ISO 10993-1** before any skin contact[11]
- **EN 13726** simulated exudate for tests[10]
- **IEC 60825-1** laser work during fabrication

Roadmap

FUTURE SCOPE · GATED PLAN

4-Month Plan on the programme's own dates: each gate must pass first



Build & Bench (Months 1 - 2)

- **Nov:** E2 laser tests (G1); build and bench-test the reader (G2).
- **Early Dec:** first strips on a dressing; 3-min prototype video.

Validate & Demo (Months 3 - 4)

- **Dec:** strip v0.2 + printed clamp; E4 wet tests (G3).
- **Jan:** repeatability; virtual semi-final with a live demo.
- **Stage 3:** OK → CHANGE live on a dressing at IIT Indore.

Technical Challenges & Risks

FOUND IN OUR OWN VERIFICATION

Geometry: Seal Under Clamp

Found: seal tape overlapped the clamp (9.0 mm³).

Impact: would lift the strip off the pins.

How: automated clash check.

Circuit: Hidden Lead Break

Found: a cracked sense lead reads 'dry'.

Impact: CHANGE never shown, no alarm.

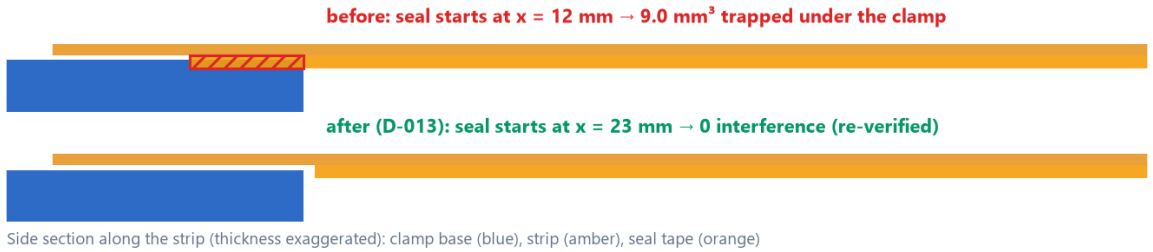
How: design FMEA (RPN 224).

Process: Unknown Laser Recipe

Found: '12' is likely a % command; tubes are unsteady at low power[8].

Impact: every strip size is provisional.

How: LaserCAD config audit.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



Also Caught: STEP exported in cm (now mm); a drawing showed 0.13 for 0.125 mm (fixed).

Potential Solutions & Risk Mitigation

FIXED · DESIGNED · FUTURE

Fixed & Verified

Seal: starts at x = 23 mm; 0 clash.

Units: all STEP files in mm.

Checks: 99 CAD checks pass.

Designed, to Prove

v0.2 strip: every sense lead looped.

Contact: silver pads, rounded pins.

Self-test: check resistor → FAULT.

Fallbacks

Wider lines: if E2 says so.

Process: slower passes, log current.

Concept: one-zone strip (score 4.29).

Future Scope & Validation Plan

- **Experiments:** E2 laser, reader bench test, E4 wet tests.
- **Material:** Raman and SEM before any 'graphene' claim.
- **Reliability:** bending, contact cycling, humidity drift.
- **Product:** battery pod; phone app on the BLE broadcast; strips per dressing.
- **Before patients:** ISO 10993, IEC 60601-1, ethics approval.
- **Not claimed:** sensing performance, graphene, clinical benefit.

Rule We Keep: no performance, graphene or clinical claim until it is measured.

References & Sources

SOURCES WITH READING DEPTH

Numbers match the superscripts on each slide. After each source we state how deeply it was read; nothing is labelled verified beyond that.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read

2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts

3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract

4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read

5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage

6. Dealey C., Cameron J., Arrowsmith M., A study comparing two objective methods of quantifying the production of wound exudate, J. Wound Care 15(4):149–153 (2006) · abstract (methods disagree)

7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read

8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed

9. Li Y. et al., Laser-induced graphene in controlled atmospheres: from superhydrophilic to superhydrophobic surfaces, Adv. Mater. 29:1700496 (2017). doi:10.1002/adma.201700496 · abstract

10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network

11. DuPont position statement on the use of DuPont materials in medical applications (Kapton caution) · web page

12. Texas Instruments ADS1115 datasheet (SBAS444); Espressif ESP32-C3 datasheet · vendor documents

13. Wounds UK news (22 Aug 2018) on a Mölnlycke-commissioned survey of >200 UK patients with chronic wounds: average dressing time 18 min · patient-reported, industry-sponsored; web page read

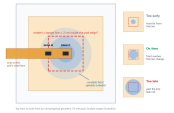
14. Smith+Nephew, ALLEVYN LIFE product page: visual change indicator ('At 75%, it's time to change your dressing') · manufacturer web page, read 19 Oct 2026

15. Truemeds.in retail listings: Mepilex Lite 10 × 10 cm ₹242; Mepilex Border 10 × 10 cm ₹514 (pack size unclear) · retail pages, read 19 Oct 2026

Image Sources

ALL IMAGES ORIGINAL

Every image in this deck was produced by the team from our own project files. No stock photos, no AI-generated images and no external images are used.



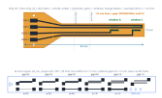
Problem illustration, drawn to scale
Source: make_figures.py · strip_v0_1_geometry.json
1_presentation/figures/fig_problem.png



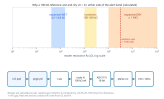
After-state intended workflow
Source: make_figures.py · firmware states
1_presentation/figures/fig_journey_after.png



Firmware scan and state logic
Source: make_figures.py · reader_v0_1 source
1_presentation/figures/fig_dataflow.png



Strip and E2 coupon layers
Source: make_figures.py · our DXF files
1_presentation/figures/fig_strip.png



Measurement window
Source: make_figures.py · calc_readout.py
1_presentation/figures/fig_window.png



Decision matrix + iterations
Source: make_figures.py · decision_matrix.md
1_presentation/figures/fig_decision.png



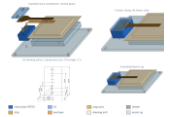
4-month roadmap
Source: make_figures.py · E2/bench/E4 plans
1_presentation/figures/fig_roadmap.png



Before-state pain-point map
Source: make_figures.py · cited sources 1–3
1_presentation/figures/fig_journey_before.png



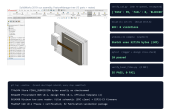
System block diagram
Source: make_figures.py · verification logs
1_presentation/figures/fig_block.png



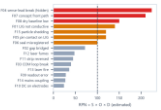
SolidWorks renders + drawing
Source: render_cad.py · 5_cad/solidworks (our models)
1_presentation/figures/fig_cad.png



KiCad schematic export
Source: kicad-cli · reader_v0_1.kicad_sch (ours)
1_presentation/figures/fig_schematic.png



SolidWorks screenshot + logs
Source: screen capture of our machine · verbatim tool output
1_presentation/figures/fig_maker.png



FMEA risk ranking
Source: make_figures.py · fmea_v0_1.csv
1_presentation/figures/fig_fmea.png



Seal clash + lead-break diagrams
Source: make_figures.py · geometry + FMEA
1_presentation/figures/fig_challenges.png